UNIVERSITY OF PETROLEUM AND ENERGY STUDIES, DEHRADUN

PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System

Software Requirements Specification Report of the (Major Project - 2) in Semester VIII

S. No	Name	Roll Number	Sap ID
1.	ANSHUMAN AGARWAL	R2142221543	500125828
2.	ADIT JAIN	R2142221275	500101867
3.	PARITOSH DHINGRA	R2142220747	500107374
4.	MUKESH SIHAG	R2142221075	500108266

BACHELOR OF TECHNOLOGY, COMPUTER SCIENCE ENGINEERING

With specialization in AI&ML

Under the guidance of

M/s Vinamrata Rani

School of Computer Science (SOCS), AIML Cluster,
UPES Bidholi Campus, Energy Acres, Dehradun – 248007

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1. INTRODUCTION

The rapid growth of urban populations and the expansion of transportation systems have significantly increased the complexity of pedestrian movement in modern cities. With more people using public spaces such as roads, transportation hubs, shopping centers, and public events, ensuring pedestrian safety and efficient crowd management has become a critical challenge. Traditional surveillance systems are mainly designed to record video footage or perform basic object detection, but they lack the ability to understand the behavior and context of pedestrian movement.

Recent advancements in artificial intelligence, particularly in computer vision and deep learning, have opened new opportunities for intelligent monitoring systems. These technologies enable machines to analyze visual data, identify objects, track movement patterns, and recognize activities in real time. Such capabilities are essential for developing advanced pedestrian monitoring solutions that can enhance situational awareness and support decision-making in smart city environments.

The PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System is designed to address these challenges by combining multiple advanced technologies into a single integrated framework. The system uses deep learning algorithms to detect pedestrians, track their movement across video frames, analyze their behavior using pose estimation techniques, and extract demographic insights such as age group and gender. By transforming raw video data into meaningful information, the system can assist authorities, security agencies, and organizations in improving safety, efficiency, and operational intelligence.

This project aims to build a scalable and intelligent pedestrian monitoring system that can be deployed in a variety of real-world environments such as traffic intersections, public surveillance systems, healthcare facilities, and retail spaces. The integration of detection, tracking, and behavioral analysis makes the system capable of providing deeper insights than traditional monitoring solutions.

1.1 Purpose of the Project

The main purpose of the PedestriGuard AI project is to design and implement an intelligent system capable of detecting, tracking, and analyzing pedestrian behavior in real time. The system aims to overcome the limitations of conventional surveillance systems that only provide visual monitoring without meaningful interpretation of pedestrian activities.

One of the key objectives of the project is to enable accurate pedestrian detection even in complex environments where challenges such as crowd density, occlusion, and dynamic movement are present. By incorporating advanced deep learning models, the system can identify pedestrians with high accuracy and track their movements continuously across video frames.

Another important purpose of the project is to analyze pedestrian behavior and identify potentially critical situations such as sudden falls or unusual movement patterns. This feature is particularly useful in healthcare monitoring systems, elderly care environments, and high-risk public areas.

In addition, the system is designed to provide demographic insights such as age group and gender estimation. These insights can be used by organizations and authorities to better understand pedestrian patterns and improve decision-making in areas such as traffic management, security planning, and retail analytics.

Overall, the project aims to demonstrate how artificial intelligence can be applied to create intelligent surveillance systems that enhance safety, efficiency, and situational awareness.

1.2 Target Beneficiary

The PedestriGuard AI system can benefit a wide range of stakeholders across different sectors. One of the primary beneficiaries is smart city management authorities, who require advanced monitoring systems to manage pedestrian traffic and ensure public safety. By analyzing pedestrian movement patterns and identifying potential risks, the system can assist in improving urban planning and traffic control.

Another important group of beneficiaries includes security and surveillance agencies responsible for monitoring public spaces such as airports, railway stations, shopping malls, and large event venues. The system can help security personnel detect unusual behavior, monitor crowd density, and respond quickly to emergency situations.

The healthcare sector, particularly elderly care facilities and hospitals, can also benefit from the system's fall detection capabilities. By monitoring patients or elderly individuals through camera systems, the technology can automatically detect accidents and trigger alerts for immediate assistance.

Additionally, retail businesses can use the system for customer analytics. By analyzing demographic information and footfall patterns, retailers can gain valuable insights into customer behavior, which can help improve marketing strategies, store layout design, and resource management.

Finally, the system can also support autonomous vehicle developers and transportation systems by providing accurate pedestrian detection and movement analysis, which are essential for safe navigation in urban environments.

1.3 Project Scope

The scope of the PedestriGuard AI project includes the development of a real-time pedestrian monitoring system that integrates multiple artificial intelligence technologies into a unified platform. The project focuses on implementing advanced algorithms for pedestrian detection, multi-object tracking, pose estimation, behavior analysis, and demographic estimation.

The system is designed to process video data from surveillance cameras or recorded video streams. It will detect pedestrians present in each video frame and track their movements across consecutive frames using tracking algorithms. Pose estimation models will analyze the body posture of pedestrians to recognize activities such as walking, running, standing, or falling.

Another important aspect within the scope of this project is the extraction of demographic insights such as age group and gender estimation from detected pedestrian images. These insights provide additional contextual information that can be used for analytical purposes.

Although the project focuses primarily on pedestrian monitoring, the design is flexible and scalable, allowing future integration with other intelligent systems such as smart traffic management systems, emergency response systems, and large-scale surveillance infrastructures.

The scope of the project also includes performance optimization techniques to ensure real-time processing capability. Model optimization tools such as ONNX and TensorRT can be used to improve inference speed and enable deployment on edge devices or GPU-enabled systems.

1.4 References

The development of PedestriGuard AI is based on extensive research and analysis of existing work in the fields of computer vision, machine learning, and intelligent surveillance systems. Several academic publications, research papers, and open-source frameworks have been studied to design the system architecture and select appropriate algorithms.

Key references include research on object detection using the YOLO family of algorithms, multi-object tracking using DeepSORT, pose estimation techniques such as OpenPose and HRNet, and fall detection systems based on deep learning models. These research contributions provide valuable insights into the design of robust pedestrian monitoring systems.

In addition to academic literature, publicly available datasets such as COCO, CityPersons, and MOTChallenge have been used for training and evaluating the models. These datasets contain annotated images and videos that represent real-world pedestrian scenarios in various environments.

The project also utilizes open-source deep learning frameworks such as PyTorch and TensorFlow, which provide powerful tools for model development, training, and deployment.

2. PROJECT DESCRIPTION

The PedestriGuard AI system is designed as a modular and scalable framework that integrates multiple deep learning components to analyze pedestrian activity in video streams. The overall architecture consists of several interconnected modules, each responsible for a specific function such as detection, tracking, behavior analysis, and demographic estimation.

The process begins with video input obtained from surveillance cameras or recorded video footage. Each video frame is analyzed using a pedestrian detection model that identifies the location of pedestrians within the frame. Once pedestrians are detected, a tracking algorithm assigns unique identities to each individual and tracks their movement across multiple frames.

Pose estimation models are then applied to detect keypoints of the human body, such as shoulders, elbows, knees, and ankles. These keypoints help determine the posture and movement patterns of pedestrians. Sequence learning models analyze the pose information over time to classify activities and detect events such as falls.

Simultaneously, a demographic estimation module analyzes the visual features of the detected pedestrians to estimate attributes such as age group and gender. All these modules work together to provide a comprehensive understanding of pedestrian behavior and activity.

2.1 Reference Algorithm

The system incorporates several advanced algorithms that have proven effective in computer vision applications. For pedestrian detection, the YOLOv8 algorithm is used because of its high accuracy and ability to perform object detection in real time.

For tracking detected pedestrians, the DeepSORT algorithm is used. This algorithm combines motion prediction with deep appearance features to maintain consistent identities for pedestrians across multiple frames.

For pose estimation, the system utilizes HRNet, which is known for producing high-resolution representations of human body keypoints. These keypoints are further analyzed using sequence-based models such as LSTM or Transformer networks to recognize activities and detect abnormal events.

For demographic estimation, a convolutional neural network is used to analyze facial or body features and predict age group and gender categories.

2.2 Characteristics of Data

The system uses several publicly available datasets to train and validate the machine learning models. These datasets contain thousands of annotated images and video frames featuring pedestrians in various environments.

Before training the models, the data undergoes several preprocessing steps. Images are resized to standard dimensions, normalized to ensure consistent pixel values, and augmented with techniques such as rotation, flipping, and brightness adjustment. These preprocessing steps help improve the generalization capability of the models.

The dataset is organized into structured formats consisting of training data, validation data, and testing data. Each image or video frame is associated with annotations that specify bounding boxes, labels, and keypoints for pedestrians.

2.3 SWOT Analysis

A SWOT analysis helps in understanding the strengths, weaknesses, opportunities, and threats associated with the PedestriGuard AI system. This analysis provides insights into both the internal capabilities of the system and the external factors that may influence its deployment and adoption.

Strengths:

One of the major strengths of the PedestriGuard AI system is its ability to integrate multiple advanced technologies such as object detection, multi-object tracking, pose estimation, and demographic analysis into a single unified framework. This integration allows the system to provide a deeper understanding of pedestrian activity compared to traditional surveillance systems.

Another strength of the system is its real-time processing capability. By utilizing efficient deep learning models such as YOLOv8 and optimized inference frameworks, the system can analyze video streams and generate results with minimal delay. This makes it suitable for applications that require immediate response, such as public safety monitoring and emergency detection.

The system is also scalable and modular. Each module can be improved or replaced independently without affecting the entire system. This modular design allows the system to adapt to new technological advancements in the field of artificial intelligence.

Weaknesses:

Despite its advanced capabilities, the system has certain limitations. One of the primary weaknesses is the high computational requirement for training and running deep learning models. Real-time video processing and multi-model integration require significant processing power, especially when dealing with high-resolution video streams.

Another limitation is the dependency on high-quality datasets for training the models. If the training data does not represent real-world conditions accurately, the system's performance may decrease in complex environments such as crowded areas or low-light conditions.

Opportunities:

The demand for intelligent monitoring systems is growing rapidly due to the expansion of smart city initiatives around the world. PedestriGuard AI has the potential to be integrated into these smart infrastructures to enhance urban safety and improve traffic management.

Another opportunity lies in healthcare monitoring systems. The fall detection capability of the system can be used in elderly care facilities and hospitals to monitor patients and provide immediate alerts in case of accidents.
The system can also be extended for retail analytics and customer behavior analysis. By analyzing demographic patterns and movement trends, businesses can gain valuable insights into consumer behavior.

Threats:
One of the major threats to intelligent surveillance systems is the concern regarding privacy and data protection. The use of video monitoring systems may raise ethical issues related to personal data collection and surveillance.
Another challenge is the rapid evolution of technology. New algorithms and models are constantly being developed, which means that existing systems may require continuous updates to remain competitive and effective.

2.4 Project Features

The PedestriGuard AI system offers several key features that make it a powerful and intelligent pedestrian monitoring solution.

One of the primary features of the system is real-time pedestrian detection. Using advanced deep learning models, the system can accurately identify pedestrians in video frames and mark their locations with bounding boxes.

Another important feature is multi-object tracking. Once pedestrians are detected, the system assigns unique identification numbers to each individual and tracks their movement across multiple frames. This ensures that the same person can be continuously monitored even as they move through the camera view.

The system also includes behavior analysis capabilities. By analyzing body posture and movement patterns using pose estimation techniques, the system can recognize activities such as walking, running, or standing. It can also detect abnormal events such as falls.

Another innovative feature is demographic analysis. The system can estimate attributes such as age group and gender for detected pedestrians. These insights can be useful for analytics in areas such as retail management and public planning.

Finally, the system is designed to operate effectively in crowded and complex environments. Techniques such as occlusion handling and attention mechanisms help maintain detection accuracy even when pedestrians partially block each other.

2.5 User Classes and Characteristics

The PedestriGuard AI system is designed to serve multiple types of users with different levels of technical expertise and responsibilities.

One of the primary user classes is security personnel who monitor surveillance systems in public spaces. These users require a simple and intuitive interface that allows them to observe pedestrian activity, receive alerts, and respond quickly to unusual events.

Another group of users includes city administrators and urban planners. These users are more interested in analytical insights generated by the system, such as pedestrian movement patterns and crowd density trends.

The system can also be used by healthcare professionals, particularly in elderly care environments. In this case, the system monitors individuals and alerts caregivers if a fall or unusual behavior is detected.

Another user group consists of data analysts and researchers who may use the collected data for studying pedestrian behavior patterns or improving the system's algorithms.
Each user class has different requirements in terms of system interaction, data access, and reporting capabilities.

2.6 Design and Implementation Constraints

During the development and deployment of the PedestriGuard AI system, several constraints must be considered.

One of the major constraints is hardware limitations. Real-time video analysis requires significant computational resources, including GPUs or high-performance processors. Systems with limited hardware capabilities may experience reduced performance.

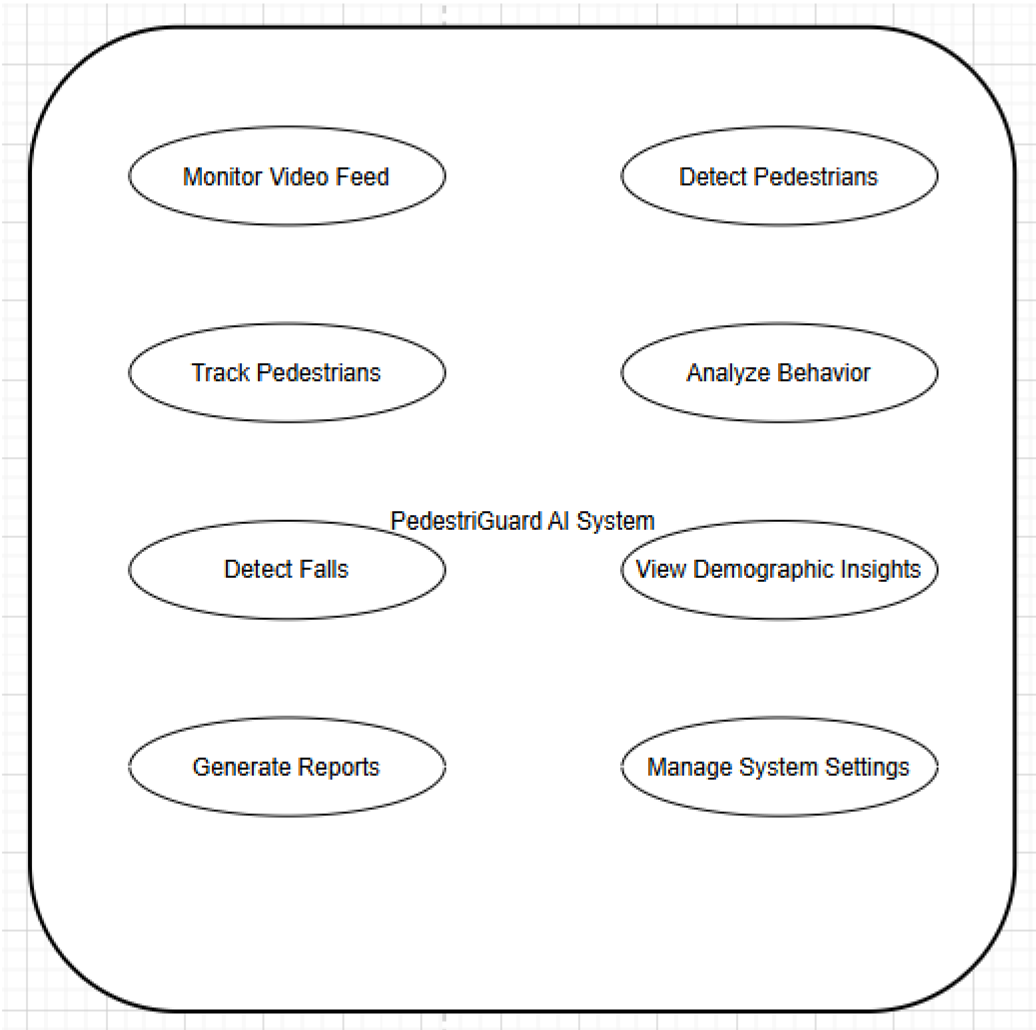
Another constraint involves data availability and quality. Training deep learning models requires large datasets with accurate annotations. Insufficient or poorly labeled data can negatively affect model accuracy.

Privacy and legal regulations also represent an important constraint. Surveillance systems must comply with data protection laws and ensure that personal data is handled responsibly.

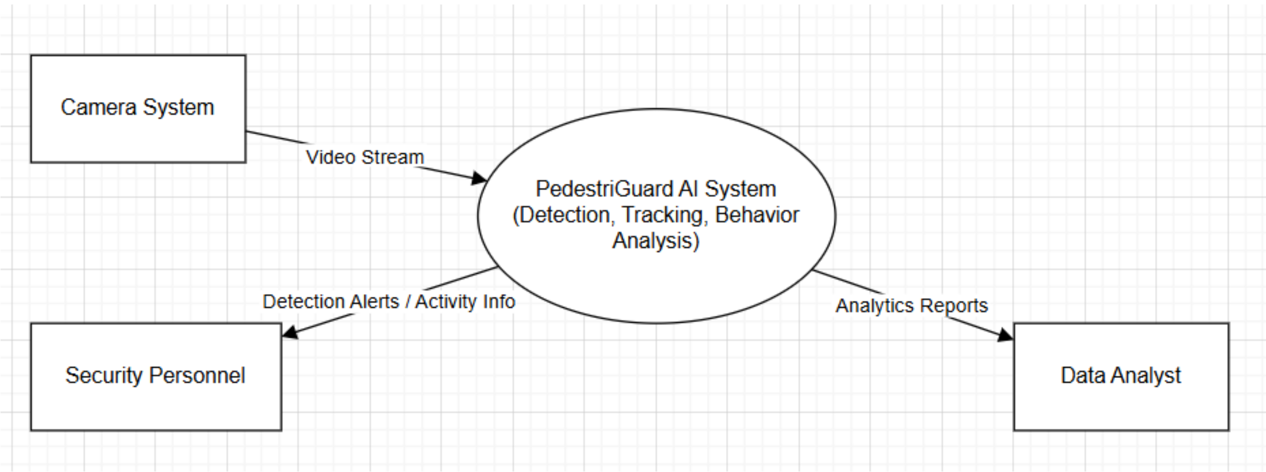
Additionally, the system must be designed to work with existing surveillance infrastructure. Compatibility with various camera systems and video formats is necessary for practical deployment.

2.7 Design Diagrams

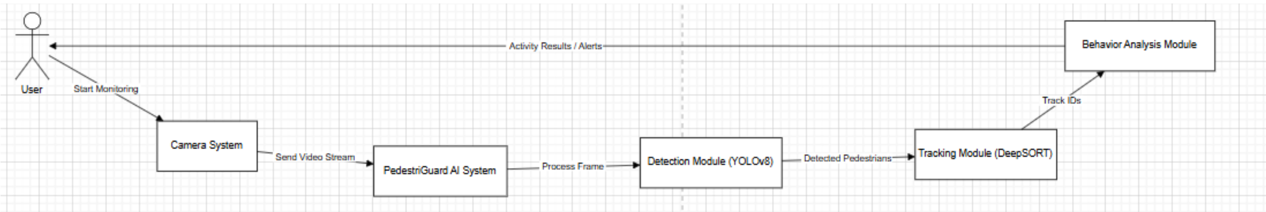
- Use Case Diagram:



- Data Flow Diagram:



- Sequence Diagram:



2.8 Assumptions and Dependencies

The development of the PedestriGuard AI system is based on several assumptions. It is assumed that the video input provided by cameras will have sufficient resolution and quality to allow accurate detection and analysis of pedestrians.

Another assumption is that the datasets used for training represent a wide variety of real-world scenarios. This ensures that the trained models can generalize well to different environments.

The system also assumes the availability of computational resources such as GPUs for training and inference tasks.

Dependencies include software frameworks such as Python, PyTorch or TensorFlow, and computer vision libraries such as OpenCV. The system also depends on external datasets and pre-trained models to accelerate the development process.

3. SYSTEM REQUIREMENTS

System requirements define the technical environment and interfaces necessary for the successful operation of the PedestriGuard AI system. These requirements specify how the system interacts with users, software components, databases, and communication protocols. Properly defining system requirements ensures that the developed system functions efficiently, integrates with existing infrastructure, and provides reliable results.

The PedestriGuard AI system processes video streams in real time, analyzes pedestrian movement using deep learning models, and generates insights such as behavior classification and demographic estimation. To support these functionalities, the system requires a well-defined user interface, compatible software frameworks, reliable data storage mechanisms, and secure communication protocols.

3.1 User Interface

The user interface (UI) of the PedestriGuard AI system is designed to be simple, intuitive, and efficient so that users from different professional backgrounds can interact with the system without requiring advanced technical knowledge.

The interface provides a dashboard view that displays live video feeds from surveillance cameras or uploaded video recordings. Detected pedestrians are highlighted using bounding boxes, and each detected individual is assigned a unique tracking ID. The system also displays activity labels such as walking, running, standing, or falling, which are generated through behavior analysis modules.

In addition to activity labels, the interface may show estimated demographic information such as age group and gender. These details appear alongside the tracked pedestrian to provide contextual information.

The user interface may also include additional features such as:

- Real-time alert notifications for abnormal events such as fall detection.
- Options to start or stop video monitoring.
- Controls for adjusting system parameters such as detection thresholds.

- Data visualization tools for analyzing pedestrian movement statistics and demographic trends.

The interface can be implemented as a desktop application, web-based dashboard, or integrated monitoring system depending on deployment requirements.

3.2 Software Interface

The PedestriGuard AI system relies on several software frameworks and libraries that enable efficient implementation of computer vision and deep learning functionalities.

The system is primarily developed using the Python programming language due to its extensive ecosystem for machine learning and data analysis. Python provides flexibility and compatibility with numerous open-source libraries that are widely used in the AI community.

Deep learning models are implemented using frameworks such as PyTorch or TensorFlow, which provide powerful tools for building, training, and deploying neural networks. These frameworks support GPU acceleration, allowing faster processing of large datasets and real-time video streams.

For image and video processing, the system uses OpenCV, a widely used computer vision library. OpenCV enables tasks such as frame extraction, image preprocessing, object visualization, and video stream handling.

Additional libraries such as NumPy, Pandas, and Matplotlib may be used for data manipulation, analysis, and visualization.

The integration of these software tools ensures that the system is capable of handling complex computer vision tasks efficiently while remaining flexible for future improvements.

3.3 Database Interface

The database interface is responsible for storing and managing the data generated by the PedestriGuard AI system. Although the primary processing occurs in real time, certain information may need to be stored for future analysis, reporting, or system evaluation.

The system may store data such as:

- Detection logs containing timestamps and pedestrian IDs
- Activity recognition results
- Demographic estimation statistics
- System performance metrics

A lightweight database such as SQLite may be used for local storage in small-scale deployments. For large-scale implementations such as smart city infrastructure, cloud-based databases such as MySQL, PostgreSQL, or MongoDB may be utilized to handle large volumes of data.

The database interface should ensure efficient data retrieval and secure storage to maintain system performance and protect sensitive information.

3.4 Protocols

Communication protocols play a crucial role in enabling the PedestriGuard AI system to interact with cameras, servers, and external systems.

Video input from surveillance cameras may be accessed using standard streaming protocols such as RTSP (Real Time Streaming Protocol) or HTTP-based streaming protocols. These protocols allow the system to capture live video feeds from IP cameras or network-based surveillance systems.

For communication between different system components, internal data transfer protocols are used to ensure efficient and secure data exchange. In distributed systems or cloud-based deployments, secure communication protocols such as HTTPS or encrypted API calls may be used to protect data integrity.

Using standardized protocols ensures that the system can be easily integrated with existing surveillance infrastructure and external monitoring systems.

4. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes and operational characteristics of the PedestriGuard AI system. Unlike functional requirements, which describe what the system does, non-functional requirements describe how well the system performs its tasks.

These requirements focus on aspects such as performance, security, reliability, scalability, and usability. Ensuring that these attributes are properly addressed is essential for deploying the system in real-world environments where consistent performance and data protection are critical.

4.1 Performance Requirements

Performance requirements specify the expected speed, responsiveness, and efficiency of the PedestriGuard AI system. Since the system processes real-time video streams, it must be capable of analyzing frames quickly and generating outputs with minimal delay.

The system should ideally process video frames at a speed close to real-time processing, which may range from 20 to 30 frames per second depending on hardware capabilities. This ensures that pedestrian detection, tracking, and behavior analysis occur without noticeable lag.

Efficient model optimization techniques such as model pruning, quantization, and hardware acceleration may be used to improve inference speed. Additionally, frameworks such as ONNX and TensorRT can be used to optimize deep learning models for faster deployment on GPUs or edge devices.

The system should also be capable of handling multiple video streams simultaneously without significant degradation in performance.

4.2 Security Requirements

Security is a critical consideration for surveillance systems that process sensitive visual data. The PedestriGuard AI system must implement appropriate security mechanisms to protect data from unauthorized access and misuse.

Access to the system should be restricted through authentication and authorization mechanisms to ensure that only authorized users can view or modify system data. User credentials and access logs should be securely managed to prevent unauthorized system access.

In cases where data is transmitted over networks, encrypted communication protocols should be used to protect data from interception. Sensitive information stored in databases should also be encrypted to ensure privacy protection.

Compliance with data protection regulations and ethical guidelines is also essential, especially when deploying the system in public environments.

4.3 Software Quality Attributes

Software quality attributes describe the overall reliability and maintainability of the system.

One important attribute is reliability, which ensures that the system consistently produces accurate detection and analysis results under different operating conditions. The system should be capable of operating continuously without unexpected failures.

Another important attribute is scalability. As surveillance networks expand, the system should be able to handle additional video streams and processing requirements without significant redesign.

The system should also be maintainable and modular. Each module of the system, such as detection, tracking, or behavior analysis, should be designed as an independent component. This modular design allows developers to update or replace specific modules without affecting the entire system.

Finally, the system should be user-friendly, ensuring that users can easily interact with the interface and understand the generated results.

5. OTHER REQUIREMENTS

In addition to functional and non-functional requirements, several additional considerations must be addressed to ensure the long-term success and usability of the PedestriGuard AI system.

One important requirement is system scalability. The system should be designed in a way that allows future expansion and integration with other intelligent systems. For example, the system may later be integrated with smart traffic management systems, emergency response systems, or large-scale surveillance networks.

Another requirement is system compatibility. The system should be compatible with different types of surveillance cameras, video formats, and operating environments. This ensures that the system can be deployed across various infrastructures without significant modifications.

The system should also support future upgrades and improvements. As new computer vision algorithms and deep learning models are developed, the system architecture should allow easy integration of improved models.

Finally, the system must address ethical and privacy considerations associated with surveillance technologies. Proper policies and safeguards should be implemented to ensure that the system is used responsibly and does not violate individual privacy rights.

Appendix A: GLOSSARY

Term	Definition
Artificial Intelligence (AI)	Artificial Intelligence refers to the simulation of human intelligence in machines that are programmed to think, learn, and perform tasks such as decision making and pattern recognition. In this project, AI enables the system to analyze video data and understand pedestrian behavior.
Computer Vision	Computer Vision is a field of artificial intelligence that allows computers to interpret and analyze visual information from images and videos. It is used in this project to detect pedestrians, analyze movement patterns, and recognize activities.
Deep Learning	Deep Learning is a subset of machine learning that uses neural networks with multiple layers to learn complex patterns from large datasets. It plays a crucial role in training the models used for detection, tracking, and behavior analysis.
Object Detection	Object detection is a computer vision technique used to identify and locate objects within images or video frames. In PedestriGuard AI, this technique is used to detect pedestrians in real-time video streams.
Multi-Object Tracking	Multi-object tracking refers to the process of following multiple objects across video frames while maintaining their unique identities. This allows the system to continuously monitor individual pedestrians.
Pose Estimation	Pose estimation is a technique used to identify key points on the human body, such as joints and limbs, to understand body posture and movement.

	It helps the system analyze pedestrian behavior and detect activities.
Behavior Analysis	Behavior analysis involves studying movement patterns and activities of pedestrians to classify actions such as walking, running, standing, or falling.
Fall Detection	Fall detection is the process of identifying sudden or abnormal movements that indicate a person has fallen. This feature is particularly useful in healthcare and elderly monitoring applications.
Convolutional Neural Network (CNN)	A CNN is a deep learning model commonly used for image analysis. In this project, CNN models help in extracting visual features and performing demographic estimation.
YOLO (You Only Look Once)	YOLO is a real-time object detection algorithm that can identify multiple objects in a single image quickly and accurately. In this project, YOLOv8 is used for detecting pedestrians in video frames.
DeepSORT	DeepSORT is a tracking algorithm that combines motion prediction and visual features to track multiple objects across frames. It helps maintain consistent identity for each detected pedestrian.
HRNet	HRNet is a high-resolution neural network used for accurate human pose estimation by detecting body keypoints.
LSTM	Long Short-Term Memory (LSTM) is a type of recurrent neural network that can learn patterns from sequential data such as movement over time. It is used for activity recognition.
ONNX	Open Neural Network Exchange (ONNX) is a format used to convert deep learning models so they can run efficiently across different frameworks and hardware platforms.
TensorRT	TensorRT is an optimization framework used to improve the speed and efficiency of deep learning models during real-time inference.

Appendix B: ANALYSIS MODEL

Component	Description
Video Input Module	This module captures video data from surveillance cameras or stored video files. The system processes each frame of the video to detect pedestrians and analyze their activities.
Frame Processing	The video stream is divided into individual frames so that each frame can be analyzed separately using computer vision algorithms. This allows the system to perform real-time processing.
Pedestrian Detection	In this stage, the YOLOv8 deep learning model analyzes each video frame and identifies the presence of pedestrians. Bounding boxes are drawn around detected individuals.
Tracking Module	After pedestrians are detected, the DeepSORT algorithm assigns unique IDs to each person and tracks them across multiple frames. This ensures consistent identification of individuals over time.
Pose Estimation Module	This module uses models such as HRNet to identify body keypoints of detected pedestrians. These keypoints help determine body posture and movement patterns.

Behavior Analysis Module	Pose data is analyzed using sequence learning models such as LSTM or Transformer networks to recognize activities like walking, running, standing, or falling.
Demographic Estimation Module	This component analyzes facial and body features using CNN models to estimate demographic attributes such as age group and gender.
Occlusion Handling	When pedestrians overlap or partially block each other, attention mechanisms and tracking logic are used to maintain detection accuracy.
Output Visualization	The final results are displayed on the user interface with bounding boxes, tracking IDs, activity labels, and demographic insights.
Data Storage	Detection results and analytics data may be stored in a database for further analysis and reporting.

Appendix C: ISSUES LIST

Issue	Description	Possible Solution
Occlusion in Crowded Environments	When pedestrians overlap or block each other in crowded areas, detection accuracy may decrease.	Implement advanced tracking algorithms and attention-based models to improve detection in crowded scenes.
Low Lighting Conditions	Poor lighting or night-time environments may reduce the accuracy of detection models.	Use image enhancement techniques and train models on diverse lighting conditions.
High Computational Requirements	Deep learning models require significant processing power for real-time inference.	Use GPU acceleration and optimize models using ONNX or TensorRT.
Dataset Bias	If the training dataset does not represent diverse demographics, the model may produce biased predictions.	Use diverse and balanced datasets during training.
Privacy Concerns	Surveillance systems may raise ethical issues related to personal privacy and data usage.	Implement strict data protection policies and ensure compliance with privacy regulations.
False Positives or False Negatives	The system may occasionally misclassify activities or fail to detect pedestrians correctly.	Improve model training with larger datasets and continuous system evaluation.
Hardware Limitations	Deployment on low-resource devices may reduce system performance.	Optimize models and use lightweight architectures for edge devices.
Integration with Existing Systems	Compatibility issues may arise when integrating the system with existing surveillance infrastructure.	Use standard communication protocols and modular system architecture.